CANDIDATE'S NAME:…….....

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Index number							

456/1

Mathematics

Paper 1

2026

2¼ hours

Apex Examination Council

PRE-ENTRY EXAMINATIONS SET TWO 2026

Uganda certificate of education

MATHEMATIS

Paper 1

2 hours 15 minutes

Apex Examination Council

INSTRUCTIONS TO STUDENTS:

- ✓ This paper has two Sections; A and B. It consists of six examination items.
- ✓ Section A has two compulsory items.
- ✓ Section B has two parts; I and II. Answer one item from each part.
- ✓ Answer four examination items in all.
- ✓ Any additional item(s) answered will not be scored.
- ✓ All answers must be written in the answer booklet(s) provided.
- ✓ Begin each item on a fresh page.
- ✓ Graph paper is provided.
- ✓ Silent non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

SECTION A

Answer all item in this Section.

Item 1.

Mr. Raban has three children he plans to visit on Saturday. He wants to give them some pocket money such that the first born gets 40% of the money, the second born to get $0.\overline{24}$ and the remainder to be given to the youngest child. The second born revealed to you that he got Ugx. 24,000. Mr. Raban also plans to construct a goat shed for his goats, the base should be rectangular to fit in the corner allocated by his family. The corner is of area $2.5m2$ and the dimensions of the goat shed are $128cm$ by $(256)\frac{3}{4}cm$.

Task.

(a) How much money did the first and last children get?

(b) Determine the simplified area of the goat shed in centimeters in form of $\boldsymbol{an}$.

Based on your calculations, will the goat shed fit in the corner?

Item 2.

Musa bought 25 text books and 35 exercise books at Ugx. 135,000 from Kampala modern styles bookshop. From the same bookshop, Esther bought 21 text books and 38 exercise books and spent Ugx. 13,000 less than Musa's expenditure.

In Mukono bookshop, the cost of a text book is 5% less and that of exercise book is 5% more than in Kampala Modern styles book shop. Esther still bought the same number of text books and exercise books and even Musa from Mukono bookshop.

Task.

(a) As a S.3 student, help Musa and Esther to find out the price of an exercise book and text book in Kampala Modern styles book shop.

(b) Which book shop would you advise Musa and Esther to buy from and why?

SECTION B

This Section has two parts; I and II Part I

Answer only one item from this part.

Item 3.

A poultry farm has three units: Sunrise, Meadow and Harvest. Each unit produces eggs and broilers, with the following monthly output.

Sunrise: 75 trays of eggs and 50 broilers

Meadow: 60 trays of eggs and 54 broilers.

Harvest: 50 trays of eggs and 32 broilers.

The farmer sells each tray of eggs for 8,500 shillings and each broiler for 10,000 shillings. To optimize the parking process, the farm owner decides to weigh a sample of eggs and clarify them according to their mass (in grams).

The table below shows the eggs and their masses.

Mass in grams	35-39	40-44	45-49	50-54	55-59	60-64	65-69
Number of eggs	20	90	180	110	40	50	10

Mr. Kirabo plans to pack eggs in different masses to meet some customer requests.

Task.

(a) Determine the sales he made after selling the farm outputs.

(b) Using a suitable statistical graph, determine the mass of majority of the eggs in the farm.

(c) Find the probability that a given pack of eggs weighs less than 50 grams.

Mr. Kirabo plans to pack eggs in different masses to meet some customer requests.

Item 4.

A city's transport department is working on improving traffic and public transport for a busy area where 124 factory workers commute daily. They have conducted a survey to understand the preferred modes of transport used by the workers. The findings have revealed that 32 workers use buses, 50 use motorcycles and 72 walk to work. Some workers use more than one mode of transport. 15 use both buses and walk, 18 use both buses and motorcycles, and 7 use both motorcycles and walk. 4 workers use only buses, while the number of workers who use none of the three transport modes (they might work remotely or use personal cars) exceeds those who use both motorcycles and walk by 3. The transport planners claim that they will put traffic lights only if the chance of those who only walk exceeds 40%. The transport planners want to analyze this data to optimize road usage and reduce congestion especially during peak hours.

Task.

(a) Determine the number of workers who use all use all the three modes of transport.

(b) Calculate the probability that a randomly selected worker uses at least two of the modes of transport.

(c) Advise the transport department, based on your calculations, whether to put traffic lights.

Part II

Answer only one item from this part.

Item 5.

Three people Amon, Brian and Charles are village friends. They decided to go hunting for wild animals in a village forest. They left their bottle of water at point A(2, 4) and their spears at point B(-1, 1) and started tracing the wild animals from point C(5, 3). While exploring the area the next day, they discovered a mysterious circular rock enclosing the area covered by three points just touching the tips of the three points.

Every 5 kilograms of meat of a wild animal costs Ugx. 30,000. A discount of 10% is given to a buyer who buys every 20 kilograms of meat. Joseph needs to buy 100 kilograms but doesn't know how much he will pay. He approaches you for help.

Task.

(a) Using a graph and suitable geometric instruments, show the design of the area covered by a circular rock.

(b) Work out the area of the circular rock.

(c) Help joseph to know how much he will pay for the meat.

Item 6.

Your father is one of the organizers of a marathon. They want to draw the map of the route that the participants will use during the race.

At their chosen starting point, they chose to take a road that turned E30^{0}S and they moved for 5km where they set up point B which will be used as a check point. They then turned through 235^0 moving a distance of 9km to point C which will be the finishing point. However, on returning to the office, they decided that the finishing point should be put at point A to cut the costs of organizing the two places but they were not sure of the details of that route from C to A that had to be included on the map.

They want to hire a vehicle that will be used to film the racers. The vehicle available consumes 2 litres per kilometer and a litre costs Ugx. 5,500. The owner of the vehicle has asked for Ugx. 500,000 plus fueling the vehicle for the total distance to be covered but the vehicle owner plans to buy fuel from the fuel station where he is a given a discount of 5% for every 100,000 worth of fuel he buys since he is a regular customer.

Task.

(a) Help your father determine the direction from point C to the new finishing point A that will be shown on the map to be drawn.

(b) Determine the total cost of hiring the vehicle.

(c) Do you think the vehicle owner will save some money on fuel? Give a reason for your answer. **END**